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H2 Mathematics (9758)
2026

2026 H2 Math Vectors I Basics Worksheet: Solutions

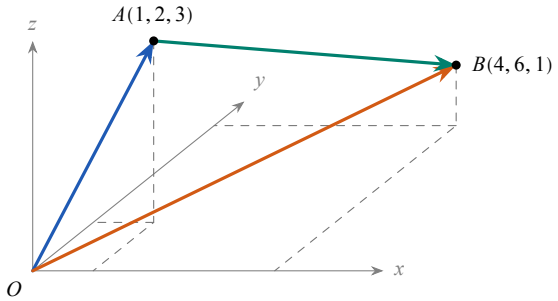
Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product. Total: 29 marks (Practice Problems; sections 1–8 are unmarked exercises).

1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a}=3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$ as a column vector.

Solution		Comments
	$\mathbf{a}=\begin{pmatrix}3\\-2\\5\end{pmatrix}$	

2 The points A and B have coordinates $(1,2,3)$ and $(4,6,1)$. Find $\overrightarrow{AB}$.



Solution		Comments
	$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$ $=\begin{pmatrix}4\\6\\1\end{pmatrix}-\begin{pmatrix}1\\2\\3\end{pmatrix}$ $=\begin{pmatrix}3\\4\\-2\end{pmatrix}$	

3 Given $\mathbf{p}=2\mathbf{i}+\mathbf{k}$ and $\mathbf{q}=-\mathbf{i}+3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p}-3\mathbf{q}+\mathbf{r}=\mathbf{0}$.

Solution		Comments
	$\mathbf{p}=\begin{pmatrix}2\\0\\1\end{pmatrix}, \quad \mathbf{q}=\begin{pmatrix}-1\\3\\0\end{pmatrix}$ $\mathbf{r}=3\mathbf{q}-2\mathbf{p}$ $=3\begin{pmatrix}-1\\3\\0\end{pmatrix}-2\begin{pmatrix}2\\0\\1\end{pmatrix}$ $=\begin{pmatrix}-3\\9\\0\end{pmatrix}-\begin{pmatrix}4\\0\\2\end{pmatrix}$ $=\begin{pmatrix}-7\\9\\-2\end{pmatrix}$	

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$	**T**
(b)	The **j**-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1 $4\mathbf{i}-\mathbf{j}+2\mathbf{k}=\begin{pmatrix}4\\-1\\2\end{pmatrix}$	**T**
(c)	The position vector of $P(1,-2,0)$ is $\mathbf{i}-2\mathbf{j}$ $\overrightarrow{OP}=\begin{pmatrix}1\\-2\\0\end{pmatrix}=\mathbf{i}-2\mathbf{j}$	**T**
(d)	$3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$ $3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\0\\2\end{pmatrix}\neq\begin{pmatrix}3\\2\\0\end{pmatrix}$	**F**
(e)	$\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$ $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$	**F**

2 Magnitude

1 Find $\left|\begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}\right|$.

Solution		Comments
	$\left\|\begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}\right\| = \sqrt{2^2+(-3)^2+6^2}$ $= \sqrt{49}$ $= 7$	

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

Solution		Comments
	$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$ $AB = \left\|\overrightarrow{AB}\right\| = \sqrt{2^2+(-6)^2+3^2}$ $= \sqrt{49}$ $= 7$	

3 Given $\mathbf{a} = \begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}| = 3$, find the possible values of t.

Solution		Comments
	$\lvert\mathbf{a}\rvert^2 = t^2 + 2^2 + (t-1)^2$ $9 = t^2 + 4 + t^2 - 2t + 1$ $0 = 2t^2 - 2t - 4$ $0 = t^2 - t - 2$ $0 = (t-2)(t+1)$ $t = 2$ or $t = -1$	

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\lvert-\mathbf{a}\rvert=\lvert\mathbf{a}\rvert$ for every vector $\mathbf{a}$ $\lvert-\mathbf{a}\rvert=\lvert-1\rvert\,\lvert\mathbf{a}\rvert=\lvert\mathbf{a}\rvert$	**T**
(b)	$\lvert3\mathbf{a}\rvert=3\lvert\mathbf{a}\rvert$ $\lvert3\mathbf{a}\rvert=\lvert3\rvert\,\lvert\mathbf{a}\rvert=3\lvert\mathbf{a}\rvert$	**T**
(c)	$\left\lvert\begin{pmatrix}0\\-5\\0\end{pmatrix}\right\rvert=-5$ $\left\lvert\begin{pmatrix}0\\-5\\0\end{pmatrix}\right\rvert=\sqrt{0^2+(-5)^2+0^2}=5$	**F**
(d)	$\left\lvert\begin{pmatrix}1\\1\\1\end{pmatrix}\right\rvert=1$ $\left\lvert\begin{pmatrix}1\\1\\1\end{pmatrix}\right\rvert=\sqrt{1^2+1^2+1^2}=\sqrt{3}\neq1$	**F**
(e)	$\left\lvert\begin{pmatrix}1\\2\\-2\end{pmatrix}\right\rvert=1+2-2=1$ $\left\lvert\begin{pmatrix}1\\2\\-2\end{pmatrix}\right\rvert=\sqrt{1^2+2^2+(-2)^2}=3$	**F**
(f)	$\lvert\mathbf{a}+\mathbf{b}\rvert=\lvert\mathbf{a}\rvert+\lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$ $\lvert\mathbf{i}+\mathbf{j}\rvert=\sqrt{2}\neq\lvert\mathbf{i}\rvert+\lvert\mathbf{j}\rvert=2$	**F**
(g)	$\lvert\mathbf{a}+\mathbf{b}\rvert\leq\lvert\mathbf{a}\rvert+\lvert\mathbf{b}\rvert$ for all vectors $\mathbf{a}$, $\mathbf{b}$ $\overrightarrow{OA}=\mathbf{a},\ \overrightarrow{AB}=\mathbf{b},\ \overrightarrow{OB}=\mathbf{a}+\mathbf{b},\ OB\leq OA+AB\ (\triangle OAB)$	**T**
(h)	For all $\lambda\in\mathbb{R}$, $\lvert\lambda\mathbf{a}\rvert=\lambda\lvert\mathbf{a}\rvert$ $\lambda=-1,\ \mathbf{a}=\mathbf{i}:\ \lvert-\mathbf{i}\rvert=1\neq-1;\ \lvert\lambda\mathbf{a}\rvert=\lvert\lambda\rvert\lvert\mathbf{a}\rvert$	**F**

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

Solution		Comments
	$\left\lvert\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}\right\rvert = \sqrt{2^2+(-1)^2+2^2}$ $= 3$ $\text{unit vector} = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$	

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

Solution		Comments
	$\left\lvert\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}\right\rvert = \sqrt{3^2+0^2+(-4)^2}$ $= 5$ $\text{vector} = 10 \cdot \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$ $= \begin{pmatrix} 6 \\ 0 \\ -8 \end{pmatrix}$	

3 The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

Solution		Comments
	$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$ $\left\lvert\overrightarrow{AB}\right\rvert = \sqrt{2^2 + (-6)^2 + 3^2}$ $= 7$ $\text{vector} = -14 \cdot \frac{1}{7} \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} -4 \\ 12 \\ -6 \end{pmatrix}$	

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector $\left\lvert\begin{pmatrix}1\\1\\1\end{pmatrix}\right\rvert = \sqrt{3} \neq 1$	**F**
(b)	$\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector $\left\lvert\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}\right\rvert = \frac{1}{3}\sqrt{1+4+4} = 1$	**T**
(c)	$\mathbf{i}+\mathbf{j}$ is a unit vector $\lvert\mathbf{i}+\mathbf{j}\rvert = \sqrt{1^2+1^2+0^2} = \sqrt{2} \neq 1$	**F**
(d)	$\lvert\hat{\mathbf{a}}\rvert = 1$ for every non-zero vector $\mathbf{a}$ $\lvert\hat{\mathbf{a}}\rvert = \left\lvert\frac{\mathbf{a}}{\lvert\mathbf{a}\rvert}\right\rvert = \frac{\lvert\mathbf{a}\rvert}{\lvert\mathbf{a}\rvert} = 1$	**T**
(e)	$\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel $\mathbf{a} = \lvert\mathbf{a}\rvert\,\hat{\mathbf{a}}$	**T**

4 Parallel Vectors ($\mathbf{a}=\lambda\mathbf{b}$)

1 Show that $\mathbf{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}-6\\3\\-9\end{pmatrix}$ are parallel.

Solution		Comments
	$\mathbf{b}=\begin{pmatrix}-6\\3\\-9\end{pmatrix}$ $=-3\begin{pmatrix}2\\-1\\3\end{pmatrix}$ $=-3\mathbf{a}$ $\therefore \mathbf{a}\parallel\mathbf{b}$ (shown)	

2 Find p and q such that $\begin{pmatrix}4\\p\\-2\end{pmatrix}$ is parallel to $\begin{pmatrix}-2\\3\\q\end{pmatrix}$.

Solution		Comments
	$\begin{pmatrix}4\\p\\-2\end{pmatrix}=\lambda\begin{pmatrix}-2\\3\\q\end{pmatrix}$ $4=-2\lambda$ $\lambda=-2$ $p=3\lambda$ $=-6$ $-2=\lambda q$ $-2=-2q$ $q=1$	

3 Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

Solution		Comments
	$\|\mathbf{a}\| = \sqrt{1^2+2^2+(-2)^2}$ $= 3$ $\text{vector} = \pm 6 \cdot \frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ $= \pm\begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$	

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$ $2\mathbf{i} \parallel \mathbf{i},\ 2\mathbf{i} \neq \mathbf{i}$	**F**
(b)	$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$ $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix} = -2\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$	**T**
(c)	$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ $\frac{2}{1} = \frac{4}{2} \neq \frac{6}{4}$	**F**
(d)	$\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$ $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix} = -3\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$	**T**
(e)	If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions $\mathbf{a} = \lambda\mathbf{b},\ \lambda = -2$	**F**

5 Collinear Points

1 Show that the points $A(1,2,3)$, $B(3,5,4)$ and $C(7,11,6)$ are collinear.

Solution		Comments
	$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 7 \\ 11 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix}$ $= \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix}$ $= 2\overrightarrow{AB}$ $\therefore \overrightarrow{AB} \parallel \overrightarrow{BC},\ B$ common $\therefore A, B, C$ collinear (shown)	

2 For the points in Question 1, find the ratio $AB : BC$.

Solution		Comments
	$\overrightarrow{BC} = 2\overrightarrow{AB}$ $BC = 2AB$ $AB : BC = 1 : 2$	

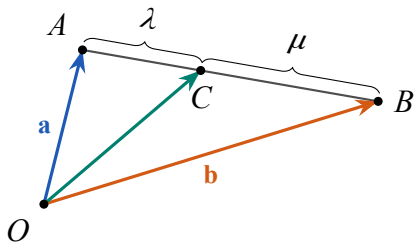
3 The points $P(2,-1,0)$, $Q(4,1,3)$ and $R(a,7,b)$ are collinear. Find a and b.

Solution		Comments
	$\overrightarrow{PR} = \lambda \overrightarrow{PQ}$ $\begin{pmatrix} a-2 \\ 8 \\ b \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$ $8 = 2\lambda$ $\lambda = 4$ $a-2 = 2(4)$ $a = 10$ $b = 3(4)$ $= 12$	

4 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear $A(0,0,0), B(1,0,0), C(0,1,0), D(1,1,0) : \overrightarrow{AB} = \overrightarrow{CD}$	**F**
(b)	If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear $\overrightarrow{AB} \parallel \overrightarrow{AC}$, A common	**T**
(c)	$A(0,0,0)$, $B(1,1,1)$ and $C(2,2,3)$ are collinear $\overrightarrow{AC} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \neq \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$	**F**
(d)	The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$) $2\mathbf{a} - \mathbf{a} = \mathbf{a}$, $-\mathbf{a} - \mathbf{a} = -2(\mathbf{a})$	**T**

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC:CB=\lambda:\mu$, $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$.

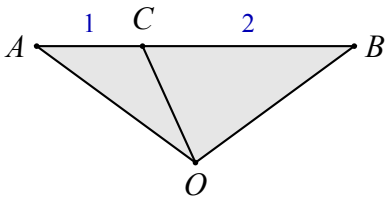
1 Let $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

(a) $AC:CB=1:2$,

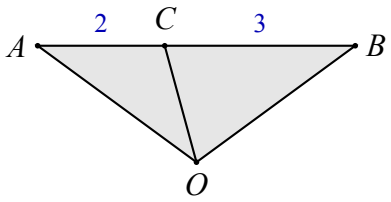
(b) $3AC=2CB$,

(c) $2AB=5CB$,

(d) Given instead that C lies on AB produced, and $3AC=7BC$.

Solution		**Comments**
(a)	 $\overrightarrow{OC}=\dfrac{2\overrightarrow{OA}+\overrightarrow{OB}}{3}$ $=\begin{pmatrix}1\\ \frac{4}{3}\\ \frac{5}{3}\end{pmatrix}$ $\therefore C\left(1,\frac{4}{3},\frac{5}{3}\right)$	

(b)



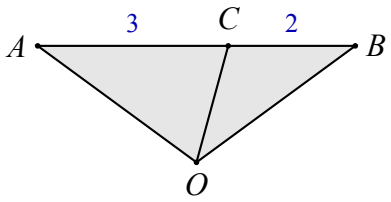
$$\frac{AC}{CB}=\frac{2}{3}$$

$$\overrightarrow{OC}=\frac{2\overrightarrow{OB}+3\overrightarrow{OA}}{5}$$

$$=\begin{pmatrix}1\\ \frac{7}{5}\\ \frac{9}{5}\end{pmatrix}$$

$\therefore C\left(1, \frac{7}{5}, \frac{9}{5}\right)$

(c)



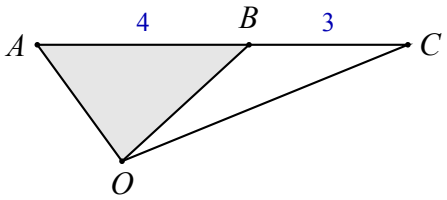
$$\frac{AB}{CB}=\frac{5}{2}$$

$$\overrightarrow{OC}=\frac{3\overrightarrow{OB}+2\overrightarrow{OA}}{5}$$

$$=\begin{pmatrix}1\\ \frac{8}{5}\\ \frac{11}{5}\end{pmatrix}$$

$\therefore C\left(1, \frac{8}{5}, \frac{11}{5}\right)$

(d)



$$\frac{AC}{BC}=\frac{7}{3}$$

$$\overrightarrow{OB}=\frac{3\overrightarrow{OA}+4\overrightarrow{OC}}{7}$$

$$\overrightarrow{OC}=\frac{7\overrightarrow{OB}-3\overrightarrow{OA}}{4}$$

$$=\begin{pmatrix}1\\ \frac{11}{4}\\ \frac{9}{2}\end{pmatrix}$$

$\therefore C\left(1, \frac{11}{4}, \frac{9}{2}\right)$

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3, -1, 5)$ and $(-1, 5, 1)$.

Solution		Comments
	$\overrightarrow{OM}=\frac{1}{2}\left[\begin{pmatrix}3\\-1\\5\end{pmatrix}+\begin{pmatrix}-1\\5\\1\end{pmatrix}\right]$ $=\frac{1}{2}\begin{pmatrix}2\\4\\6\end{pmatrix}$ $=\begin{pmatrix}1\\2\\3\end{pmatrix}$	

3 For the following statements, state whether it is **true** or **false**.

	Statement	T/F
(a)	$\overrightarrow{OC}=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$ $\overrightarrow{OC}=\mathbf{a}+\frac{\lambda}{\lambda+\mu}(\mathbf{b}-\mathbf{a})=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$	**T**
(b)	$\overrightarrow{OC}=\frac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$ $\overrightarrow{OC}=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$	**F**
(c)	If $\lambda:\mu=1:1$, then $\overrightarrow{OC}=\frac{1}{2}(\mathbf{a}+\mathbf{b})$ $\overrightarrow{OC}=\frac{1\mathbf{a}+1\mathbf{b}}{2}$	**T**
(d)	If $\lambda:\mu=1:3$, then $\overrightarrow{OC}=\frac{3\mathbf{a}+\mathbf{b}}{4}$ $\overrightarrow{OC}=\frac{3\mathbf{a}+1\mathbf{b}}{1+3}$	**T**
(e)	If $\lambda:\mu=2:1$, then $\overrightarrow{OC}=\frac{2\mathbf{a}+\mathbf{b}}{3}$ $\overrightarrow{OC}=\frac{1\mathbf{a}+2\mathbf{b}}{3}$	**F**

7 Scalar Product

1 Properties of scalar product

It is given that $\mathbf{a}=\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

(a) $\mathbf{a}\cdot\mathbf{b}$ (b) $\mathbf{b}\cdot\mathbf{a}$ (c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$ (d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$

(e) $\mathbf{a}\cdot(2\mathbf{c})$ (f) $(2\mathbf{a})\cdot\mathbf{c}$ (g) $\mathbf{a}\cdot\mathbf{a}$

Ans: (a) 22 (b) 22 (c) 13 (d) 13 (e) -18 (f) -18 (g) 14

2 Scalar product formula: $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$

(a) Points O, A, B are such that $\overrightarrow{OA}=\mathbf{i}+4\mathbf{j}-3\mathbf{k}$ and $\overrightarrow{OB}=5\mathbf{i}-\mathbf{j}$

(i) Find the size of angle AOB

(ii) Find the size of angle OAB.

(b) It is given that $\mathbf{c}=\begin{pmatrix}2\\3\\6\end{pmatrix}$, $\mathbf{d}=\begin{pmatrix}1\\-2\\3\end{pmatrix}$ and $\mathbf{e}=\begin{pmatrix}3\\2\\-2\end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and

(i) the vector $\mathbf{d}$

(ii) the vector $\mathbf{e}$

(iii) the positive y-axis

(c) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:

$$|\mathbf{a}||\mathbf{b}|=6.$$

Find the value of $\mathbf{a}\cdot\mathbf{b}$ for each of the following scenarios.

(i) The angle between $\mathbf{a}$ and $\mathbf{b}$ is 60°.

(ii) The angle between $\mathbf{a}$ and $\mathbf{b}$ is 120°.

(iii) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Ans:

(a)(i) 87.8° (ii) 46.1°

(b)(i) 57.7° (ii) 90° (iii) 64.6°

(c)(i) $\mathbf{a}\cdot\mathbf{b}=3$ (ii) $\mathbf{a}\cdot\mathbf{b}=-3$ (iii) $\mathbf{a}\cdot\mathbf{b}=0$

3 Perpendicularity of vectors

(a) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $2\mathbf{i}+4\mathbf{j}+\mathbf{k}$ respectively. Show that OA is perpendicular to OB.

(b) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a}=2p\mathbf{i}+3p\mathbf{j}+6p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$$

where $p>0$. It is given that $|\mathbf{a}|=|\mathbf{b}|$.

(i) Find the exact value of p.

(ii) Show that the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$ are perpendicular.

Ans: (b)(i) $p=\frac{3}{7}$

4 Length of projection

(a) Given that $\mathbf{a}=2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+2\mathbf{j}-\mathbf{k}$, find the length of projection and the projection vector of

(i) $\mathbf{a}$ onto $\mathbf{b}$,

(ii) $\mathbf{b}$ onto $\mathbf{a}$.

(b) With reference to the origin O, the points A, B and C have position vectors given by

$$\mathbf{a}=\mathbf{i}+2\mathbf{j}-\mathbf{k},\ \mathbf{b}=3\mathbf{i}+\mathbf{k} \text{ and } \mathbf{c}=2\mathbf{i}+2\mathbf{j}+3\mathbf{k}$$

respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

Ans:

(a)(i) $\frac{2}{\sqrt{6}}, \frac{1}{3}\begin{pmatrix}1\\-2\\1\end{pmatrix}$ (ii) $\frac{2}{3}, -\frac{2}{9}\begin{pmatrix}2\\1\\2\end{pmatrix}$

(b) $\frac{2}{3}$

8 Vector Product

1 Properties of vector product

(a) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

(b) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(i) $\mathbf{a} \times \mathbf{b}$; (ii) $\mathbf{b} \times \mathbf{a}$; (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (iv) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$;

(v) $\mathbf{a} \times (2\mathbf{b})$; (vi) $(2\mathbf{a}) \times \mathbf{b}$; (vii) $\mathbf{a} \times \mathbf{a}$;

Ans:

(a) $\begin{pmatrix} -29 \\ 17 \\ 26 \end{pmatrix}$

(b)(i) $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ (ii) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ (iii) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ (iv) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ (v) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ (vi) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ (vii) $\mathbf{0}$

2 Parallel vectors

(a) Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.

(b) Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.

(i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$

(ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$

(iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

Ans:

(b)(i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = 32$, $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ -3 \end{pmatrix}$; Neither

(ii) $\mathbf{a} \cdot \mathbf{b} = 0$, $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 2 \\ -10 \\ -6 \end{pmatrix}$; Perpendicular

(iii) $\begin{pmatrix}1\\2\\-1\end{pmatrix}\cdot\begin{pmatrix}-4\\-8\\4\end{pmatrix}=-24,\ \begin{pmatrix}1\\2\\-1\end{pmatrix}\times\begin{pmatrix}-4\\-8\\4\end{pmatrix}=\mathbf{0}$; Parallel

3 Vector product formula

Using the Vector Product, find the acute angles between the following vectors:

(a) $\begin{pmatrix}2\\1\\3\end{pmatrix}$ and $\begin{pmatrix}4\\-1\\1\end{pmatrix}$ (b) $\begin{pmatrix}-1\\2\\-1\end{pmatrix}$ and $\begin{pmatrix}3\\-1\\5\end{pmatrix}$ (c) $\begin{pmatrix}4\\3\\5\end{pmatrix}$ and $\begin{pmatrix}2\\-5\\1\end{pmatrix}$

Ans: (a) 51.0° (b) 46.4° (c) 87.0°

4 Perpendicular distance

(a) Relative to an origin O, two points A and B have position vectors given by $\mathbf{a}=3\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-2\mathbf{i}+3\mathbf{j}+2\mathbf{k}$. Find the following:
  (i) The shortest distance from A to OB
  (ii) The shortest distance from B to OA

(b) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\mathbf{b}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$ respectively.
  (i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.
  (ii) Hence, or otherwise, find the shortest distance from B to OA.

(c) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$, $\mathbf{b}=\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $\mathbf{c}=3\mathbf{i}-\mathbf{j}+\mathbf{k}$. Find the shortest distance from C to the line joining A and B.

Ans:

(a)(i) 3.54 units (ii) 3.90 units

(b)(i) $2\sqrt{3}$ units (ii) $\sqrt{2}$ units

(c) 2 units

5 Areas of triangles & parallelograms

(a) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA}=\begin{pmatrix}1\\0\\3\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}-2\\3\\2\end{pmatrix}$. Find the area of triangle OAB.

(b) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA}=\begin{pmatrix}0\\1\\1\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}1\\-2\\2\end{pmatrix}$. Find the area of triangle OAB.

(c) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are

$$\mathbf{i}+\mathbf{j}-\mathbf{k},\ 2\mathbf{i}+\mathbf{k} \text{ and } -\mathbf{i}+2\mathbf{j}-\mathbf{k}$$

respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find the exact area of triangle ABC.

Ans: (a) $\frac{\sqrt{154}}{2}$ units^2 (b) $\frac{3\sqrt{2}}{2}$ units^2 (c) $\frac{\sqrt{21}}{2}$ units^2

9 Practice Problems

1 Relative to the origin O, the position vectors of two points A and B are $\mathbf{a}$ and $\mathbf{b}$ respectively. The length of $\mathbf{a}$ is 4 units and $\mathbf{b}$ is a unit vector. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{\pi}{3}$ radians.

(a) Find the area of triangle OAB. [2]

It is given that the point M lies on the line segment AB such that $AM:AB$ is $3:4$.

(b) Find the exact area of triangle OAM. [3]

It is given that the point C lies on AB produced, such that $\overrightarrow{AC}=\lambda\overrightarrow{AB}$, where $\lambda>1$.

(c) Find the value of λ such that OC is perpendicular to AB. [4]

Solution		**Comments**
(a) **[2]**	Area of $\Delta OAB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ $=\frac{1}{2}\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\frac{\pi}{3}$ $=\frac{1}{2}(4)(1)\left(\frac{\sqrt{3}}{2}\right)$ $=\sqrt{3}$ units^2	
(b) **[3]**	$AM:MB=3:1$ $\overrightarrow{OM}=\frac{1\cdot\mathbf{a}+3\cdot\mathbf{b}}{4}=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$ Area of $\Delta OAM=\frac{1}{2}\left\lvert\mathbf{a}\times\left(\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}\right)\right\rvert$ $=\frac{1}{2}\left\lvert\frac{1}{4}(\mathbf{a}\times\mathbf{a})+\frac{3}{4}(\mathbf{a}\times\mathbf{b})\right\rvert$ $=\frac{3}{4}\cdot\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert\quad(\mathbf{a}\times\mathbf{a}=\mathbf{0})$ $=\frac{3}{4}\sqrt{3}$ $=\frac{3\sqrt{3}}{4}$ units^2	
(c) **[4]**	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\frac{\pi}{3}=(4)(1)\frac{1}{2}=2$ $\overrightarrow{OC}=\mathbf{a}+\lambda(\mathbf{b}-\mathbf{a})$ $\overrightarrow{OC}\cdot\overrightarrow{AB}=0$ $[\mathbf{a}+\lambda(\mathbf{b}-\mathbf{a})]\cdot(\mathbf{b}-\mathbf{a})=0$ $\mathbf{a}\cdot\mathbf{b}-\lvert\mathbf{a}\rvert^2+\lambda(\lvert\mathbf{b}\rvert^2-2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{a}\rvert^2)=0$ $2-16+\lambda(1-4+16)=0$ $-14+13\lambda=0$ $\lambda=\frac{14}{13}$	

2 The non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy $|\mathbf{a}|=3$, $|\mathbf{b}|=4$ and $\mathbf{a}\cdot\mathbf{b}=6$.

(a) Find the angle between $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Find $|\mathbf{a}-\mathbf{b}|$. [3]

Solution		Comments
(a) **[2]**	$\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}=\dfrac{6}{3\cdot4}=\dfrac{1}{2}$ $\theta=\dfrac{\pi}{3}$ (or 60°)	
(b) **[3]**	$\lvert\mathbf{a}-\mathbf{b}\rvert^2=(\mathbf{a}-\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})$ $=\lvert\mathbf{a}\rvert^2-2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2$ $=9-2(6)+16$ $=13$ $\lvert\mathbf{a}-\mathbf{b}\rvert=\sqrt{13}$	

3 Relative to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $3\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel.

(a) Find the area of triangle OAC in terms of $|\mathbf{a}\times\mathbf{b}|$. [2]

(b) The point D is the midpoint of AC. Find the area of triangle OAD in terms of $|\mathbf{a}\times\mathbf{b}|$. [2]

Solution		**Comments**
(a) **[2]**	Area of $\Delta OAC = \frac{1}{2}\lvert\mathbf{a}\times(3\mathbf{b})\rvert$ $= \frac{1}{2}\cdot 3\,\lvert\mathbf{a}\times\mathbf{b}\rvert$ $= \frac{3}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$	
(b) **[2]**	$\overrightarrow{OD} = \frac{1}{2}(\mathbf{a}+3\mathbf{b})$ Area of $\Delta OAD = \frac{1}{2}\left\lvert\mathbf{a}\times\frac{1}{2}(\mathbf{a}+3\mathbf{b})\right\rvert$ $= \frac{1}{4}\lvert\mathbf{a}\times\mathbf{a}+3(\mathbf{a}\times\mathbf{b})\rvert$ $= \frac{1}{4}\cdot 3\,\lvert\mathbf{a}\times\mathbf{b}\rvert$ $= \frac{3}{4}\lvert\mathbf{a}\times\mathbf{b}\rvert$	

4 The non-zero vectors **a**, **b** and **c** satisfy $3\mathbf{a}\times\mathbf{b}=2\mathbf{c}\times\mathbf{b}$.

(a) Show that $3\mathbf{a}-2\mathbf{c}=\lambda\mathbf{b}$ for some scalar λ. [2]

(b) It is now given that $|\mathbf{a}|=2$, $|\mathbf{b}|=3$ and the angle between **a** and **b** is $\frac{\pi}{6}$. Find $|3\mathbf{a}-2\mathbf{b}|$. [3]

Solution		**Comments**
(a) **[2]**	$3\mathbf{a}\times\mathbf{b}-2\mathbf{c}\times\mathbf{b}=\mathbf{0}$ $(3\mathbf{a}-2\mathbf{c})\times\mathbf{b}=\mathbf{0}$ Since $\mathbf{b}\neq\mathbf{0}$, the vector $3\mathbf{a}-2\mathbf{c}$ is parallel to **b**. Hence $3\mathbf{a}-2\mathbf{c}=\lambda\mathbf{b}$ for some scalar λ.	
(b) **[3]**	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\frac{\pi}{6}=(2)(3)\frac{\sqrt{3}}{2}=3\sqrt{3}$ $\lvert3\mathbf{a}-2\mathbf{b}\rvert^2=(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})$ $=9\lvert\mathbf{a}\rvert^2-12\mathbf{a}\cdot\mathbf{b}+4\lvert\mathbf{b}\rvert^2$ $=9(4)-12(3\sqrt{3})+4(9)$ $=36-36\sqrt{3}+36$ $=72-36\sqrt{3}$ $\lvert3\mathbf{a}-2\mathbf{b}\rvert=\sqrt{36(2-\sqrt{3})}$ $=6\sqrt{2-\sqrt{3}}$	

5 Relative to the origin O, the points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}|=2$ and that $\mathbf{b}$ is a unit vector. The point C divides the line segment AB internally such that $AC:CB=3:2$. The point D lies on OB produced such that $\overrightarrow{OD}=3\overrightarrow{OB}$.

(a) Find $\overrightarrow{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(b) Find $\overrightarrow{OD}$ in terms of $\mathbf{b}$. [1]

(c) Given that CD is perpendicular to AB, find the value of $\mathbf{a}\cdot\mathbf{b}$. [3]

Solution		**Comments**
(a) **[2]**	$AC:CB=3:2$ $\overrightarrow{OC}=\dfrac{2\mathbf{a}+3\mathbf{b}}{5}=\dfrac{2}{5}\mathbf{a}+\dfrac{3}{5}\mathbf{b}$	
(b) **[1]**	$\overrightarrow{OD}=3\mathbf{b}$	
(c) **[3]**	$\overrightarrow{CD}=\overrightarrow{OD}-\overrightarrow{OC}=3\mathbf{b}-\left(\frac{2}{5}\mathbf{a}+\frac{3}{5}\mathbf{b}\right)=-\frac{2}{5}\mathbf{a}+\frac{12}{5}\mathbf{b}$ $\overrightarrow{AB}=\mathbf{b}-\mathbf{a}$ $\overrightarrow{CD}\cdot\overrightarrow{AB}=0$ $\left(-\frac{2}{5}\mathbf{a}+\frac{12}{5}\mathbf{b}\right)\cdot(\mathbf{b}-\mathbf{a})=0$ $-\frac{2}{5}\mathbf{a}\cdot\mathbf{b}+\frac{2}{5}\lvert\mathbf{a}\rvert^2+\frac{12}{5}\lvert\mathbf{b}\rvert^2-\frac{12}{5}\mathbf{a}\cdot\mathbf{b}=0$ $\frac{2}{5}(4)+\frac{12}{5}(1)-\left(\frac{2}{5}+\frac{12}{5}\right)\mathbf{a}\cdot\mathbf{b}=0$ $\frac{8}{5}+\frac{12}{5}-\frac{14}{5}\mathbf{a}\cdot\mathbf{b}=0$ $4=\frac{14}{5}\mathbf{a}\cdot\mathbf{b}$ $\mathbf{a}\cdot\mathbf{b}=\dfrac{10}{7}$	

END OF SOLUTIONS